

A Rank-Three Haar Phase Law for a Four-Symbol Open Walsh Gate

Route-A structural certificate C168

Abstract

We study the natural four-symbol open Walsh gate $A = F_4^* \text{diag}(1, 0, 1, 1)$ at the complete register clock $C_k = A^{\otimes k}$. Its characteristic polynomial is $x(x-1)(x^2 + ix/2 - 1/2)$, so the nonzero secular degree is exactly 3^k and the generalized zero space has dimension $4^k - 3^k$ for every register length. The two damped eigenvalues have normalized phases $u_{\pm}$ whose ratio r satisfies $r + r^{-1} = -3/2$. Since this nonintegral rational cannot be the algebraic-integer trace of a root of unity, r is non-torsion. The multiplicity-weighted phase measure is a three-step multiplicative walk, and its m th Fourier coefficient is $((1 + u_+^m + u_-^m)/3)^k$. Every fixed nonzero mode contracts strictly; hence the phase measures converge weakly to Haar measure on the circle. The one-site log modulus equals 0 with probability $1/3$ and $-\log 2/2$ with probability $2/3$. An exact mixed transform upgrades the marginal limits to a joint $N(0, (\log 2)^2/18) \otimes \text{Haar}$ law. The result is source-side and does not construct a self-adjoint or target-spectrum operator.

Keywords: open quantum maps; Walsh dynamics; tensor spectra; phase equidistribution; Haar measure; algebraic integers.

中文摘要

本文研究自然四符号开放 Walsh 门 $A = F_4^* \text{diag}(1, 0, 1, 1)$ 在完整寄存器时钟 $C_k = A^{\otimes k}$ 下的精确谱结构。单点特征多项式为 $x(x-1)(x^2 + ix/2 - 1/2)$ ，因此对任意寄存器长度，非零世俗多项式次数恰为 3^k ，广义零特征空间维数为 $4^k - 3^k$ 。两个衰减特征值的单位相位记为 $u_{\pm}$ ，其比值 r 满足 $r + r^{-1} = -3/2$ 。若 r 是单位根，则该迹应为代数整数；但非整数有理数 $-3/2$ 不可能如此，故相位比非扭结。按代数重数加权后，相位测度成为三步乘法游走，其第 m 个傅里叶系数精确等于 $((1 + u_+^m + u_-^m)/3)^k$ 。每个固定非零模态均严格收缩，从而相位测度弱收敛到圆上的 Haar 测度。单点对数模以概率 $1/3$ 取零、以概率 $2/3$ 取 $-\log 2/2$ ；精确混合变换进一步证明中心化涨落与相位联合收敛到方差为 $(\log 2)^2/18$ 的高斯分布和 Haar 测度的乘积律。结论只属于源侧开放动力学，不构造自伴算子或目标谱。
关键词：开放量子映射；Walsh 动力学；张量谱；相位等分布；Haar 测度；代数整数。

1 Frozen gate and one-site certificate

Let $F_4 = (i^{j\ell}/2)_{0 \leq j, \ell < 4}$, $P_1 = \text{diag}(1, 0, 1, 1)$, and $A = F_4^* P_1$. One application of the cyclic register map B_k is one tick; one complete cycle of exactly k ticks is $C_k = B_k^k = A^{\otimes k}$.

Lemma 1. A is diagonalizable with eigenvalues

$$0, \quad 1, \quad \lambda_+ = \frac{\sqrt{7} - i}{4}, \quad \lambda_- = \frac{-\sqrt{7} - i}{4}. \quad (1)$$

For $u_{\pm} = \sqrt{2}\lambda_{\pm}$, the ratio $r = u_+/u_- = (-3 + i\sqrt{7})/4$ is not a root of unity.

Proof. Direct determinant expansion gives

$$\chi_A(x) = x(x-1)(x^2 + ix/2 - 1/2), \quad (2)$$

with four distinct roots (1), and $|\lambda_{\pm}| = 1/\sqrt{2}$. Exact radical arithmetic gives $r + r^{-1} = -3/2$. Were r a root of unity, this sum would be an algebraic integer. A rational algebraic integer is an integer, a contradiction. $\square$

2 All-length secular and phase laws

For $a + b + c = k$, write $M_k(a, b, c) = k!/(a!b!c!)$.

Theorem 2. For every $k \geq 1$,

$$\det(I - zC_k) = \prod_{a+b+c=k} (1 - z\lambda_+^b \lambda_-^c)^{M_k(a, b, c)}. \quad (3)$$

The polynomial in (3) has degree 3^k , and the generalized zero eigenspace has dimension $4^k - 3^k$.

Proof. Tensor products of the four one-site eigenvectors form an eigenbasis. Nonzero products use the three nonzero roots; grouping words by their three contents gives (3) and the multinomial total 3^k . Every other tensor vector has a zero factor. $\square$

Let μ_k be the algebraic-multiplicity-weighted phase law of the nonzero spectrum. The same word count yields

$$\mu_k = 3^{-k} \sum_{a+b+c=k} M_k(a, b, c) \delta_{u_+^b u_-^c},$$

$$\hat{\mu}_k(m) = \left(\frac{1 + u_+^m + u_-^m}{3} \right)^k. \quad (4)$$

Theorem 3. μ_k converges weakly to normalized Haar measure on $\mathbb{T}$.

Proof. For $m \neq 0$, equality in the triangle inequality for the average in (4) would force $u_+^m = u_-^m = 1$, hence $r^m = 1$, contradicting Lemma 1. Every fixed nonzero Fourier mode therefore tends to zero. Density of trigonometric polynomials proves the claim. $\square$

The labels in (3) retain their multiplicities; distinctness across all triples is not asserted. Nor is a Fourier gap uniform in all modes claimed. At every finite k , μ_k is atomic, so its total-variation distance from continuous Haar measure is exactly one.

3 Joint modulus–phase theorem

Under the normalized nonzero spectral law, a tensor factor contributes $X = 0$ with probability $1/3$ and $X = -\log 2/2$ with probability $2/3$. Hence $\mathbb{E}X = -\log 2/3$ and $\text{Var } X = (\log 2)^2/18$. Put $Y_k = (\log |\rho| + k \log 2/3)/\sqrt{k}$.

Theorem 4. For real t and integer m ,

$$\mathbb{E}[e^{itY_k} \text{phase}(\rho)^m]$$

$$= e^{it \log 2 \sqrt{k}/3} \left(\frac{1 + e^{-it \log 2/(2\sqrt{k})} (u_+^m + u_-^m)}{3} \right)^k. \quad (5)$$

Consequently $(Y_k, \text{phase}(\rho)) \Rightarrow N(0, (\log 2)^2/18) \otimes \text{Haar}$.

Proof. For $m = 0$, (5) is the centered iid characteristic function and converges to the stated Gaussian transform. For fixed $m \neq 0$, its base tends to the strict contraction in (4), so the mixed transform tends to zero. The scalar CLT supplies tightness; Fourier–characteristic uniqueness gives the product limit. $\square$

4 Evidence and boundary

The release freezes exact ledgers through $k = 24$ and modes $m = 24$. Independent strict Python and SymPy reconstructions, byte replay, and hostile semantic mutations test the implementation; the finite ledgers do not prove the all- k statements.

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